Software Requirements Specification

for

texlab

Version 1.0 approved

Prepared by:
Aryan Rastogi (2427030083)
Raghav Sethi (2427030127)

Organization: texlab Project

Date Created: October 20, 2026

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Revision History

Name	Date	Reason For Changes	Version
Aryan Rastogi (2427030083) Raghav Sethi (2427030127)	2026-10-20	Formal Software Requirements Specification release conforming to Karl E. Wiegers template with UI screenshots, DFDs, sequence, activity, state transition, package, and object diagrams.	1.0 approved

1. Introduction

1.1 Purpose & Scope

This Software Requirements Specification (SRS) details the requirements for **texlab** (Version 1.0), an intelligent, web-based natural language to publication-grade LaTeX generator. texlab converts natural-language descriptions and sketches into valid TikZ diagrams, booktabs tables, amsmath equations, and PGFPlots charts. The platform guarantees compliance with IEEE two-column (8.5 cm) bounds via raster fit validation, offloads compilation to texlive.net without local TeX dependencies, performs autonomous multi-pass compilation repair, and enables direct exports to Overleaf and ZIP archives.

1.2 Document Conventions & Audience

Conforms to Karl E. Wiegers (1999) SRS template and IEEE Std 830-1998. Statements follow RFC 2119 keywords (SHALL, SHOULD, MAY). Functional requirements use REQ-FeatureNumber.SequenceNumber. Target audience includes frontend/UI developers, backend/cloud engineers, prompt engineers, QA testers, and academic researchers.

1.3 References

1. Wiegers, K. E., *SRS Template*, 1999. 2. IEEE Std 830-1998. 3. Tantau, T., *TikZ & PGF Manual v3.1.10*, 2023. 4. OpenRouter AI Documentation. 5. Texlive.net Latexcgi Service. 6. Overleaf API Documentation.

2. Overall Description

2.1 Product Perspective & Functions

texlab is a decoupled web application comprising a React 18 / Vite SPA and a Node.js serverless proxy layer. Compilation is executed remotely via texlive.net latexcgi (pdflatex standalone engine), and AI inference is routed via OpenRouter (gpt-4o-mini, gemini-2.5-flash/pro). Core functions: Natural language parsing, multimodal sketch ingestion (up to 4 images), keyword intent detection, preference-steered formatting, curated template catalog with parametric editing, remote compilation proxy, autonomous compiler error repair, raster-based canvas column fit validation, multi-agent workspace tabs, local version history (up to 30 figures), and multi-format export.

2.2 User Classes and Characteristics

User Class	Frequency	Technical Expertise & Key Functions	Priority
Academic Researchers	Daily/Weekly	Advanced LaTeX; limited TikZ. Requires publication figures strictly fitting IEEE 8.5 cm column margins; heavy Overleaf export.	Highest
Students & Educators	Periodic	Intermediate LaTeX; novice TikZ. Generates thesis diagrams, project timelines, and homework equations; relies on templates and auto-repair.	High
Software & ML Engineers	Frequent	Advanced software; intermediate LaTeX. Generates neural topologies, architectures, and comparison tables; utilizes sketch uploads.	High
Technical Writers	Bi-weekly	Moderate LaTeX. Converts whiteboard drawings into clean vector charts; utilizes standalone ZIP bundles and monochrome modes.	Medium

2.3 Operating Environment & Constraints

• **Operating Environment:** Modern web browsers (Chrome 110+, Firefox 115+, Safari 16.4+, Edge 110+) with Canvas/Fetch API. Serverless Node.js 18.x/20.x runtime.
• **Design Constraints:** Stateless execution (60s timeout); zero local TeX toolchains; texlive.net CORS avoidance via /api/compile-latex proxy; standalone pdflatex class only; payload limits (4,000 prompt chars, 200,000 LaTeX chars, 12 MB ZIP, 4 image attachments).
• **Documentation & Dependencies:** Repository documentation (README.md, ARCHITECTURE.md, library/STYLE.md) and template previews. External dependencies: OpenRouter, texlive.net, Upstash Redis, Overleaf API.

3. External Interface Requirements

3.1 User Interfaces

texlab implements a clean, high-contrast, responsive interface built with Tailwind CSS, Lucide icons, and Radix UI primitives. It features: (1) **Landing Page (/)** with centered hero prompt card, format triggers (Diagram, Table, Equation, Plot), multimodal image attachment, and curated template catalog carousel; (2) **Workspace Desk (/app)** featuring AI chat and iterative refinement on the left, and tabbed live PDF preview (PDF.js) and LaTeX source editor on the right.

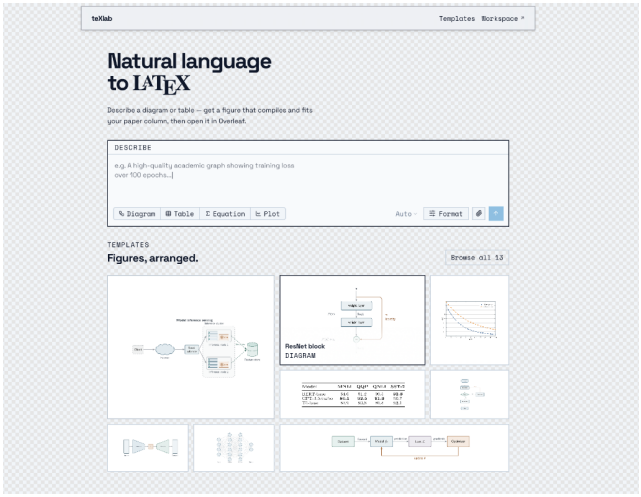


Figure 3.1: texlab Landing Page with Input Card & Template Gallery

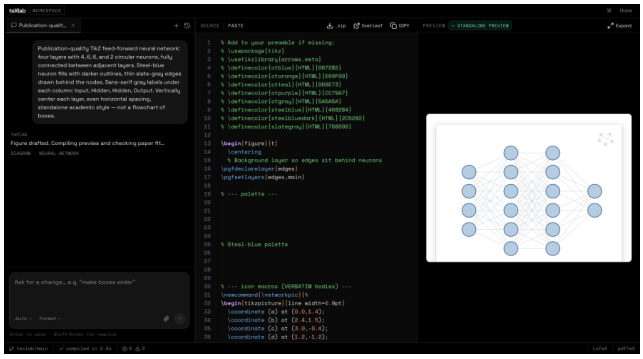


Figure 3.2: texlab Workspace Desk with Left Chat & Right Live Preview

3.2 Hardware, Software & Communication Interfaces

• **Hardware Interfaces:** Pointing devices, touch screens (canvas pan/zoom), and retina/high-DPI pixel ratio scaling.
• **Software Interfaces:** OpenRouter LLM API (HTTPS POST, Bearer auth, gpt-4o-mini, gemini-2.5-flash/pro); Texlive.net Latexcgi (multipart/form-data, pdflatex); Overleaf API (POST form with base64 snip_uri); Upstash Redis (REST pipeline API for rate limits and moderation).
• **Communications:** TLS 1.3 encryption. Security response headers: X-Content-Type-Options: nosniff, X-Frame-Options: DENY, Referrer-Policy: strict-origin-when-cross-origin, Permissions-Policy.

4. System Features

4.1 Natural Language to LaTeX & Multimodal Input — *Priority: High*

Converts text prompts and up to 4 reference sketches into valid LaTeX code. **REQ-1.1:** Accept prompts up to 4,000 characters and 4 reference images. **REQ-1.2:** Route requests with images exclusively to vision-capable models. **REQ-1.3:** Sanitize output by stripping markdown fences and commentary. **REQ-1.4:** Sanitized HTTP 502/504 errors on upstream timeouts.

4.2 Fine-Grained Generation Preferences Steering — *Priority: High*

Controls structural and aesthetic formatting options. **REQ-2.1:** Support output types (diagram, table, equation, plot) and sub-styles (flowchart, architecture, neural-net, timeline, hierarchy, comparison, ablation). **REQ-2.2:** Support palettes: academic (tlblue, tlteal, tlorange, tlpurple, tlgray), monochrome, pastel, vivid. **REQ-2.3:** Support canvas aspect ratios: auto, square (1:1 boustrophedon), landscape, portrait, wide. **REQ-2.4:** Support fits: standalone, snippet, column (8.5 cm bound), fullpage (17 cm bound).

4.3 Automated Prompt Keyword & Style Detection — *Priority: Medium*

Infers output type and sub-style from prompt keywords. **REQ-3.1:** Scored regex matching executes debounced as user types; specific domain phrases outrank generic single tokens.

4.4 Interactive Dual-Pane Workspace & Live Compilation Preview — *Priority: High*

Split desk interface for real-time authoring. **REQ-4.1:** Left pane handles conversational chat and prompt revisions. **REQ-4.2:** Right pane switches between Live PDF Preview (rendered via pdfjs-dist with 50%–200% zoom/pan) and syntax-highlighted LaTeX source editor.

4.5 Remote LaTeX Compilation Proxy & Log Capture — *Priority: High*

Proxies LaTeX source to texlive.net latexcgi without client-side toolchains. **REQ-5.1:** Wrap snippet bodies in standalone document preamble with preloaded packages (tikz, pgfplots, amsmath, booktabs, xcolor). **REQ-5.2:** Stream PDF binary (HTTP 200) on success; return TeX log text with HTTP 422 on failure within 45s timeout.

4.6 Autonomous Compile-Error Diagnostics & Multi-Pass Auto-Repair — *Priority: High*

Autonomously resolves LaTeX syntax errors. **REQ-6.1:** Intercept 422 compilation errors, extract '!'-prefixed error lines, and dispatch up to 2 auto-repair iterations with mode='repair'. **REQ-6.2:** Relocate misplaced keys, declare missing packages/colors, and fix undefined node coordinates.

4.7 Raster-Based Column Fit Validation & Auto-Fit Refinement — *Priority: High*

Inspects rasterized preview canvas bounding box pixels to verify academic column width compliance. **REQ-7.1:** Pixel scanner locates non-white bounding pixels; if content touches right margin (marginRight < 0.012), status is marked 'cropped'. **REQ-7.2:** Autonomously trigger one fit-refine pass to scale, reflow, or wrap TikZ nodes.

4.8 Curated Figure Template Library & Parametric Editing — *Priority: Medium*

Provides publication-grade templates in library/catalog.json. **REQ-8.1:** matchLibraryTemplate matches prompts to proven craft patterns for neural nets, flowcharts, architectures, and comparison tables.

4.9 Multi-Agent Session Management & Local Figure Version History — *Priority: Medium*

Multi-tab workflows without remote accounts. **REQ-9.1:** Persist up to 30 figures with version trees (latex, label, timestamps) in browser localStorage. **REQ-9.2:** Restore previous version snapshots instantly upon chip selection.

4.10 Multi-Destination Export (Clipboard, Overleaf Integration, ZIP Bundles) — *Priority: High*

Diverse export pipelines. **REQ-10.1:** Copy paste-ready snippets formatted for columnwidth or textwidth. **REQ-10.2:** Download structured ZIP containing figure.tex, snippet.tex, preamble.tex, .tex, and README.txt. **REQ-10.3:** Open projects in Overleaf via base64 data URIs (<1.5 MB) or server tokenized snip URIs.

4.11 Semantic Moderation, Security Gate, & Prompt Injection Defense — *Priority: High*

Guards against abuse and jailbreaks. **REQ-11.1:** Enforce figure-only allowlist via Llama Guard / GPT-4o-mini (cached in Redis for 600s). **REQ-11.2:** leaksSystemPrompt() inspects model outputs for canary tokens, replacing leaks with '% Request declined.'. **REQ-11.3:** Enforce sliding-window rate limiting per IP and cookie (texlab_rl).

5. Other Nonfunctional Requirements

5.1 Performance Requirements

• **NFR-PERF-1:** Remote compilation via /api/compile-latex SHALL complete within **3.0s** under nominal conditions.
• **NFR-PERF-2:** Initial code generation via OpenRouter SHALL return within **15.0s** for standard figures.
• **NFR-PERF-3:** PDF preview canvas re-rendering upon zoom/pan SHALL execute at **60 fps** with redraw latency < **50 ms**.
• **NFR-PERF-4:** Total production JavaScript transfer size for initial landing SHALL not exceed **350 KB** gzipped.

5.2 Safety Requirements

• **NFR-SAFE-1:** Block generation of actionable weapon designs, chemical/biological threats, or malware attack flows.
• **NFR-SAFE-2:** Workspace continuously serializes prompts and code to localStorage, guaranteeing zero loss across crashes.
• **NFR-SAFE-3:** Preview canvases explicitly release image buffers and revoke Blob object URLs to prevent memory leaks.

5.3 Security Requirements

• **NFR-SEC-1:** Server-side API keys (OPEN_ROUTER_API, UPSTASH_REDIS_REST_TOKEN) SHALL NEVER be transmitted to browsers.
• **NFR-SEC-2:** Sliding-window rate limits enforced per IP and cookie (texlab_rl): Generation: 8/min burst, 40/hr, 100/day; Compile: 30/min burst, 200/hr, 500/day. Exceeding returns HTTP 429 Retry-After.
• **NFR-SEC-3:** Preferences strictly validated against hardcoded string whitelists before prompt interpolation.

5.4 Software Quality Attributes & 5.5 Business Rules

• **Quality Attributes:** 99.9% uptime via Vercel Edge; first-pass compile success >= 92% (rising to >= 98% after auto-repair); automatic multi-model fallback on provider rate limits; first-time user time-to-figure < 60 seconds.
• **Business Rules:** BR-1: Strictly scoped to technical/academic LaTeX figures; non-figure chat queries rejected. BR-2: Fast models (gpt-4o-mini, gemini-2.5-flash) prioritized for drafts/repairs; heavy models (gemini-2.5-pro) reserved for complex figures. BR-3: Core generation operates without mandatory user login.

6. Other Requirements

• **6.1 Database & Caching:** Upstash Redis stores rate limit windows and SHA-256 moderation hashes (TTL: 600s). In-memory Map fallback when Redis is unconfigured.
• **6.2 Internationalization:** UTF-8 character encoding throughout; non-ASCII characters converted to compatible TeX macros or inputenc.
• **6.3 Legal & Licensing:** Distributed under the MIT License. Generated LaTeX output imposes no copyleft encumbrances on user research papers.

Appendix A: Glossary

Term	Definition
AST	Abstract Syntax Tree; structured hierarchical representation of code syntax.
Booktabs	LaTeX package providing publication-grade typographic rules for tables without vertical rules.
Latexcgi	Online compilation CGI service hosted by texlive.net converting TeX POST source into binary PDFs.
LLM	Large Language Model (e.g., GPT-4o, Gemini 2.5) providing natural language reasoning and code synthesis.
Overleaf	Collaborative cloud-based LaTeX authoring platform with direct project import capabilities.
PDF.js	Open-source HTML5/WebAssembly PDF rendering engine developed by Mozilla.
Pdflatex	Standard TeX engine compiling TeX/LaTeX directly into Adobe Portable Document Format.
PGFPlots	High-level LaTeX package built on TikZ for drawing publication-quality 2D and 3D scientific plots.
Standalone	LaTeX document class (\documentclass{standalone}) auto-clipping page size exactly to content bounding box.
TikZ	Powerful declarative vector graphics language for producing high-precision technical diagrams in TeX.

Appendix B: Analysis Models — Data Flow Diagrams

B.1 Level 0 Data Flow Diagram (DFD Level 0)

Represents texlab as a single process interacting with external entities (User, AI Model / LaTeX Compiler).

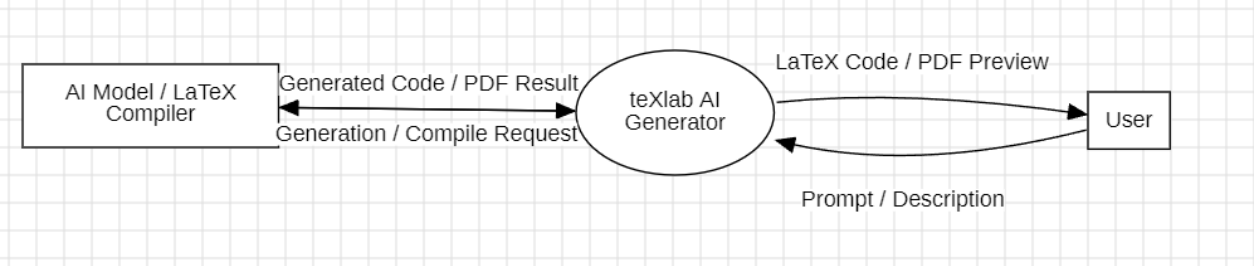


Figure B.1: Level 0 Data Flow Diagram (Context View)

B.2 Level 1 Data Flow Diagram (DFD Level 1)

Decomposes texlab into primary processes: prompt capture, code generation, compilation, and result display.

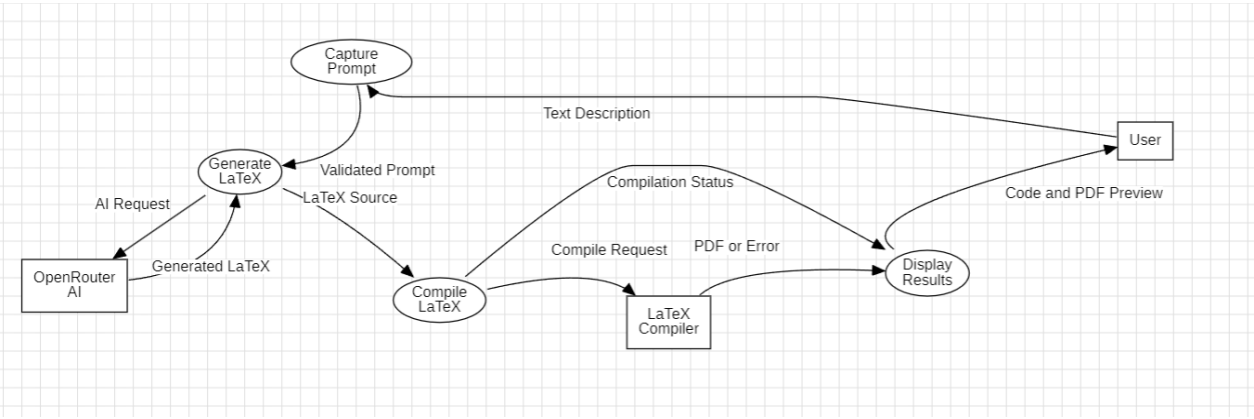


Figure B.2: Level 1 Data Flow Diagram (Decomposed Major Processes)

B.3 Level 2 Data Flow Diagram (DFD Level 2)

Detailed internal pipeline: prompt validation, AI request dispatching, code generation, compiler execution, and preview rendering.

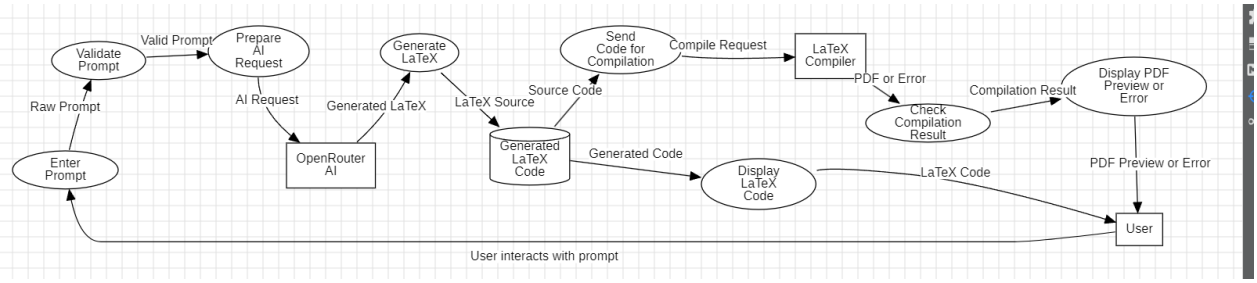


Figure B.3: Level 2 Data Flow Diagram (Detailed Internal Operations)

Appendix B: Analysis Models — Behavioral & Interaction Models

B.4 Sequence & B.5 Activity Diagrams

Chronological message flows across system actors and end-to-end execution workflow from prompt input to error recovery.

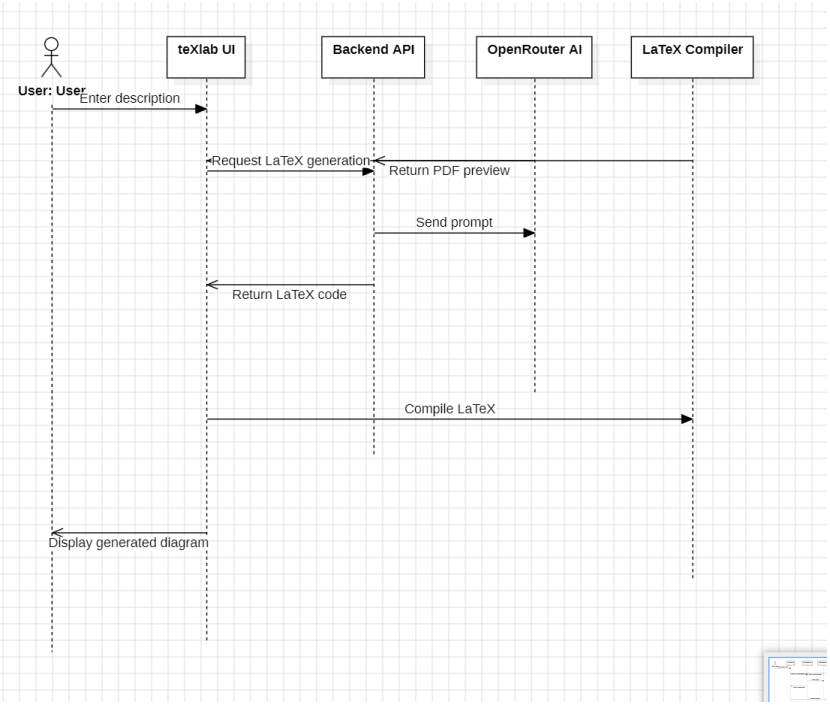


Figure B.4: Sequence Diagram for texlab Generation & Compilation Flow

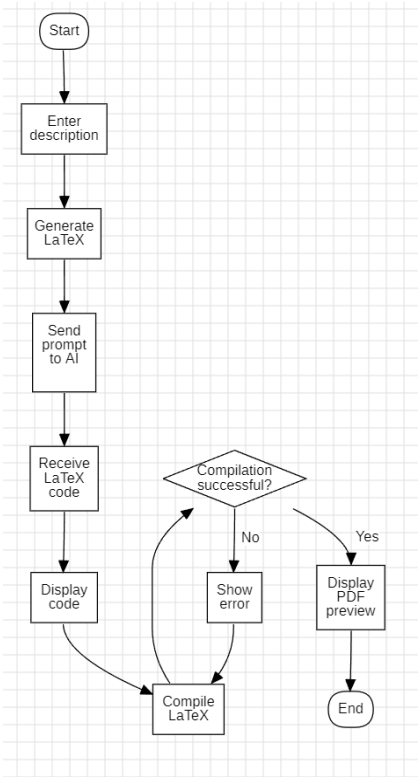


Figure B.5: Activity Diagram for Request Workflow & Error Recovery

B.6 State Transition Diagram

Lifecycle states of a generation request: Idle, PromptEntered, Validating, Generating, CodeReady, Compiling, PreviewReady, Error.

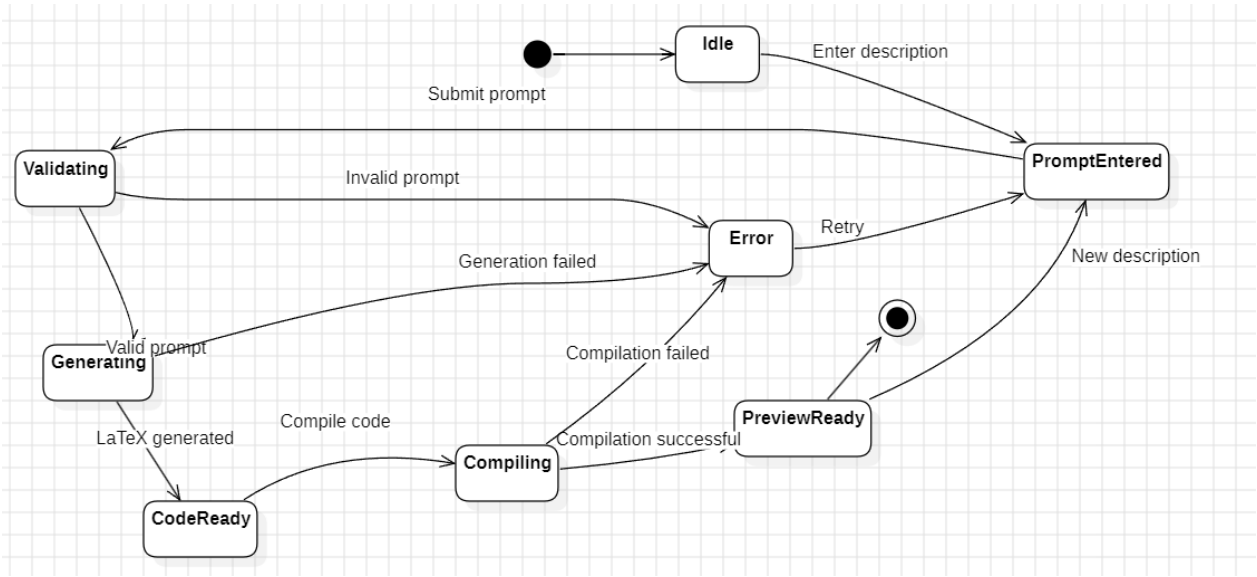


Figure B.6: State Transition Diagram for LaTeX Generation Request Lifecycle

Appendix B: Analysis Models — Architectural & Object Structure

B.7 Package Diagram

Logical package architecture illustrating module dependencies across UserInterface, PromptProcessing, AICodeGeneration, LaTeXCompilation, PreviewAndExport, and APIServices.

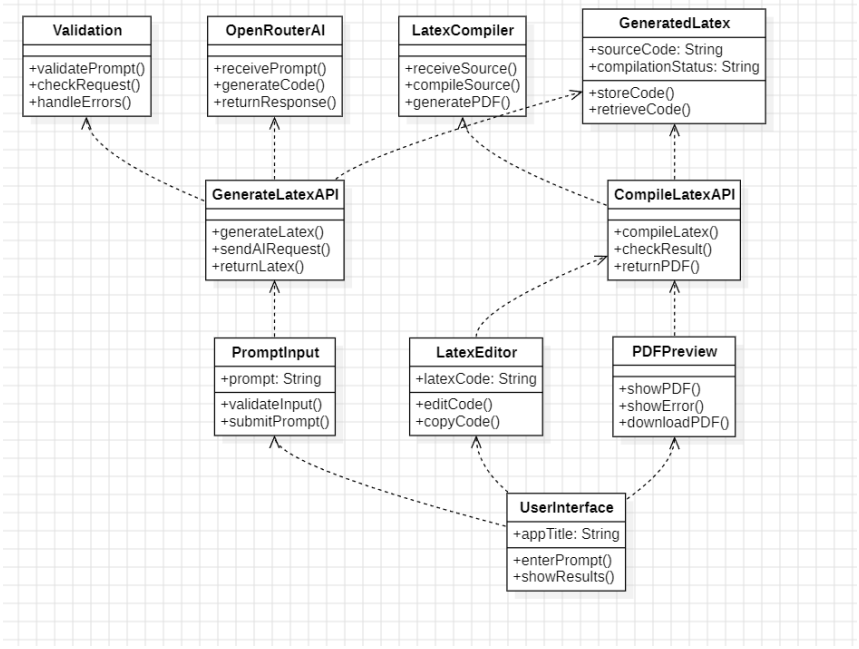


Figure B.7: Package Diagram for texlab Application Architecture

B.8 Object Diagram

Runtime object snapshot during active code generation, showing collaborating entity instances.

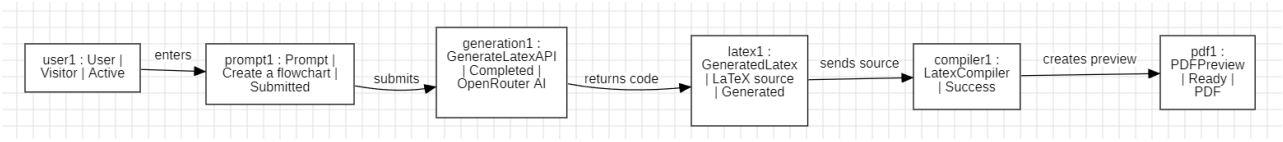


Figure B.8: Object Diagram representing Runtime System Snapshot during Generation

Appendix C: To Be Determined List

Item ID	Section	Description of Open Issue / Enhancement	Target Milestone	Owner
TBD-1	2.5 / 6.1	Implementation of an authenticated cloud database for cross-device synchronized user figure libraries beyond localStorage.	v1.2	Cloud Team
TBD-2	3.3 / 4.5	Support for additional compilation engines (xelatex, lualatex) to allow arbitrary OpenType and system font rendering.	v1.3	Backend Team
TBD-3	4.10	Direct client-side SVG vector extraction from compiled PDFs via pdf2svg WebAssembly module for HTML embed snippets.	v1.2	Frontend Team
TBD-4	5.3	Migration of all ephemeral in-memory rate-limiter fallback buckets to globally shared Redis instances across all regions.	v1.1	DevOps Team
TBD-5	4.1	Support for audio / speech-to-text prompt transcription for hands-free whiteboard-to-LaTeX figure authoring.	v2.0	AI Research